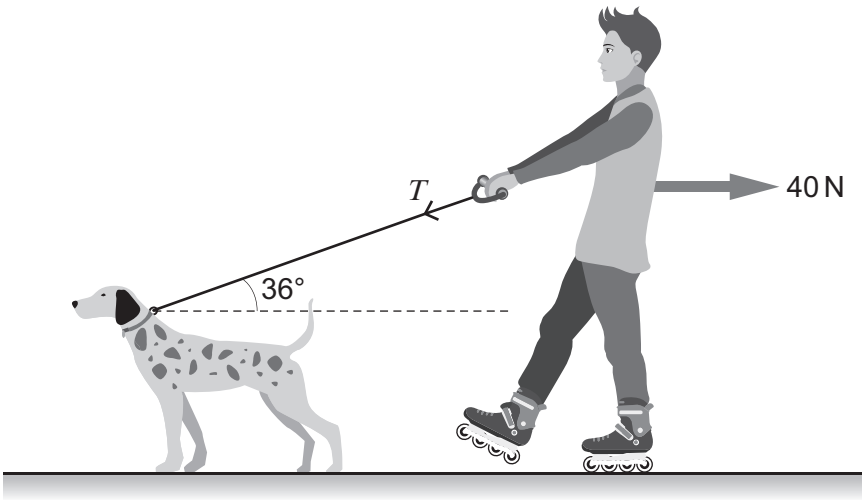


4. (a) The diagram shows a roller-skater being pulled along a horizontal road at **constant velocity**. The total resistive force to the skater's motion is 40 N.



Show that the tension, T , in the dog lead is approximately 50 N. [2]

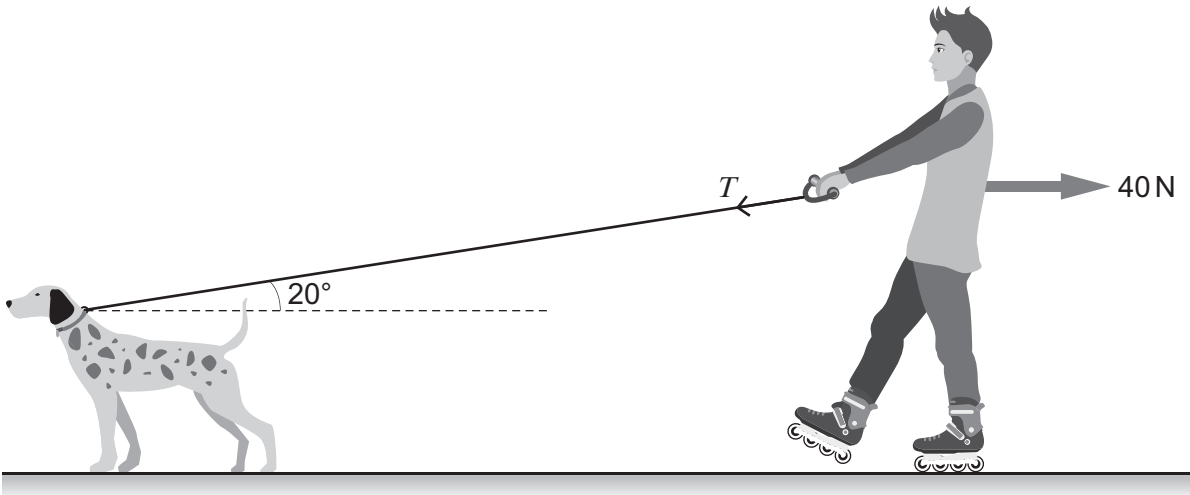
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- (b) Later, the roller-skater, moving at the same velocity as in part (a), extends the dog lead as shown in the diagram below.



- (i) Assuming that the tension, T , is the same as in part (a), explain why the roller-skater initially accelerates. [1]

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(ii) Calculate this initial acceleration, given that the mass of the roller-skater is 35 kg. [3]

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(c) The roller-skater believes that the dog’s rate of doing work is the same, regardless of the length of the dog lead. Use information from (a) and (b) to investigate whether or not this claim is correct. [3]

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